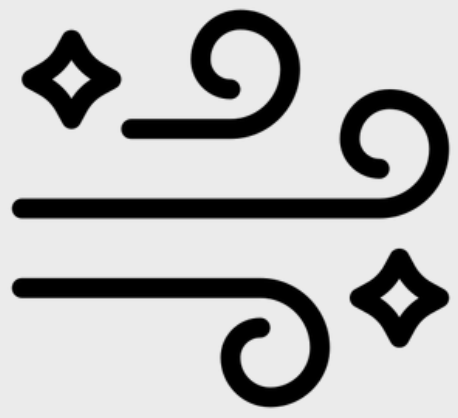




Air Quality Anaylsis



Content

Business Understanding >

Data Preparation >

Data Modeling & Evaluation >

Business Objective >

Data Visualization >

Streamlit app & Conclusion >

• Business Understanding

- **Business Problem:** Air pollution is a major environmental and public health concern.

This project focuses on analyzing air quality in Taiwan from 2016 to 2024 to understand the main factors and pollutants associated with high Air Quality Index (AQI) levels.

- **The Dataset Contains:**

1. AQI measurements
2. Pollutants such as *PM2.5, PM10, SO₂, CO, O₃, NO₂, and NO_x*
3. Weather and environmental factors
4. Station and county information
5. Geographic location
6. Date and time information

- **The project will also use Machine Learning to predict AQI based on pollutant concentrations and environmental conditions.**



• Business Objective & Solutions

• Business Objective :

1. Analyze air pollution patterns across Taiwan.
2. Identify the areas and stations with the highest pollution levels.
3. Determine which pollutants are most strongly associated with high AQI.
4. Understand how air quality changes over time.
5. Develop a Machine Learning model to predict AQI using pollutant and environmental measurements.

• Business Solutions :

1. Identify the most polluted counties and monitoring station.
2. Identify the pollutants most strongly related to high AQI.
3. Understand air quality trends from 2016 to 2024.
4. Analyze the impact of weather conditions on air quality.
5. Build a Machine Learning model that can predict AQI.
6. Evaluate the model using appropriate regression metrics:
7. Compare different models and select the Best Performing Model Based on the evaluation results.



- # Data Preparation

- **Preparing the raw dataset for reliable analysis and machine learning by:**

1. Cleaning missing, duplicate, and invalid data.
2. Correcting data types and handling outliers.
3. Creating relevant features and preparing the final dataset for modeling.



• Data Preparation

Duplicate

```
df.duplicated().sum()
```

```
np.int64(0)
```

Drop useless columns

```
df.drop(columns="unit", inplace=True)
```

Clean Column Names

```
df.columns = (  
    df.columns  
    .str.strip()  
    .str.lower()  
    .str.replace(" ", "_")  
    .str.replace(".", "_")  
)
```

Type conversion

```
df["date"] = pd.to_datetime(df["date"], errors="coerce")
```

```
numeric_columns = [  
    "aqi",  
    "so2",  
    "co",  
    "o3",  
    "pm10",  
    "pm2_5",  
    "no2",  
    "nox",  
    "no",  
    "windspeed",  
    "winddirec",  
    "so2_avg",  
    "pm2.5_avg",  
    "o3_8hr",  
    "pm10_avg",  
    "co_8hr",  
    "pm2_5_avg"  
]  
  
for col in numeric_columns:  
    if col in df.columns:  
        df[col] = pd.to_numeric(df[col], errors="coerce")
```



• Data Preparation

```
measurement_columns = [  
    "aqi",  
    "so2",  
    "co",  
    "o3",  
    "pm10",  
    "pm2_5",  
    "no2",  
    "nox",  
    "no",  
    "windspeed",  
    "winddirec",  
    "so2_avg",  
    "pm2.5_avg",  
    "o3_8hr",  
    "pm10_avg",  
    "co_8hr",  
    "pm2_5_avg"  
]  
  
for col in measurement_columns:  
    if col in df.columns:  
        df.loc[df[col] < 0, col] = np.nan
```

Missing

```
round(df.isnull().sum().sort_values(ascending=False)*100/len(df), 2)
```

...	0
pollutant	84.80
winddirec	2.87
windspeed	2.84
so2	2.09
no	1.74
no2	1.74
nox	1.74
o3	1.36
pm2_5	1.14
co	1.12
pm10	1.09
so2_avg	0.68
o3_8hr	0.53
pm2_5_avg	0.40
co_8hr	0.05



• Data Preparation

date	sitename	country	aqi	status	so2	co	o3	o3_8hr	pm10	pm2_5	no2	nox	no	windspeed	winddirec	co_8hr	pm2_5_avg
23/08/2024 9:00:00 PM	Qianjin	Kaohsiung City	22	Good	0.6	0.15	14.9	17.7	15	6	1.8	2.8	0.9	0.8	226	0.1	
23/08/2024 5:00:00 AM	Zhudong	Hsinchu County	22	Good	0.6	0.15	0.7	2.5	21	6	4	5.2	1.2	0.8	113	0.1	
22/08/2024 6:00:00 AM	Zhushan	Nantou County	19	Good	0.6	0.15	3.5	3	21	6	3.4	4.4	1	0.3	37	0.1	
21/08/2024 11:00:00 PM	Tainan (Madou)	Tainan City	18	Good	0.6	0.15	10.2	11	15	6	3.7	4.1	0.4	0.2	140	0.1	
21/08/2024 10:00:00 PM	Zhudong	Hsinchu County	18	Good	0.6	0.15	7.9	19.9	18	6	3.6	4.3	0.6	2	129	0.1	
21/08/2024 10:00:00 PM	Pingzhen	Taoyuan City	19	Good	0.6	0.15	8.7	18.3	7	6	11.8	28.5	5.3	1.7	226	0.1	
21/08/2024 8:00:00 PM	Pingzhen	Taoyuan City	25	Good	0.6	0.15	13.6	27.3	15	6	6.5	8	1.5	2.1	235	0.1	
21/08/2024 7:00:00 PM	Pingzhen	Taoyuan City	29	Good	0.6	0.15	17.6	32	13	6	5.8	7.2	1.3	2.8	244	0.1	
19/08/2024 2:00:00 PM	Shalu	Taichung City	22	Good	0.6	0.15	31.8	24.6	14	6	5.6	6.5	0.9	3.6	186	0.1	
18/08/2024 3:00:00 PM	Wanli	New Taipei City	29	Good	0.6	0.15	39.1	31	14	6	2.9	4.3	1.4	1.7	358	0.1	
17/08/2024 1:00:00 PM	Linkou	New Taipei City	23	Good	0.6	0.15	42.4	24.6	5	6	5.2	6.4	1.1	4.5	271	0.1	
17/08/2024 8:00:00 AM	Qiaotou	Kaohsiung City	23	Good	0.6	0.15	24.9	25.3	1	6	5.9	6.9	0.9	1.4	70	0.1	
17/08/2024 7:00:00 AM	Hualien	Hualien County	25	Good	0.6	0.15	6.1	9.1	29	6	1.8	2.8	1	2.9	212	0.1	
16/08/2024 7:00:00 PM	Matsu	Lienchiang County	32	Good	0.6	0.15	26	35.9	11	6	3	3.5	0.5	0.2	340	0.1	
15/08/2024 8:00:00 PM	Lunbei	Yunlin County	22	Good	0.6	0.15	20.7	24.6	8	6	4.9	5.7	0.8	0.2	148	0.1	
14/08/2024 2:00:00 PM	Nantou	Nantou County	24	Good	0.6	0.15	33.9	26.2	9	6	6.3	7.8	1.4	1.3	193	0.1	
12/08/2024 7:00:00 AM	Taitung	Taitung County	20	Good	0.6	0.15	8.4	12.4	28	6	2.3	2.9	0.5	1.5	322	0.1	
12/08/2024 7:00:00 AM	Yangming	Taipei City	26	Good	0.6	0.15	33.8	28.8	16	6	1.5	2.4	0.8	3	140	0.1	
12/08/2024 4:00:00 AM	Shalu	Taichung City	26	Good	0.6	0.15	20.8	21.7	15	6	4.5	5.1	0.5	1.8	169	0.1	
09/08/2024 10:00:00 PM	Sanyi	Miaoli County	35	Good	0.6	0.15	27.9	38.1	13	6	0.9	2.1	1.1	1.3	200	0.1	
09/08/2024 7:00:00 PM	Sanyi	Miaoli County	45	Good	0.6	0.15	37.8	49.5	5	6	0.5	1.7	1.2	3.3	176	0.1	
07/08/2024 7:00:00 AM	Taitung	Taitung County	23	Good	0.6	0.15	8.2	14.1	19	6	2.9	5	2.1	1.1	318	0.1	
03/08/2024 6:00:00 PM	Fuxing	Kaohsiung City	29	Good	0.6	0.15	22.3	26.6	17	6	4.4	5.8	1.3	0.8	259	0.1	
03/08/2024 10:00:00 AM	Meinong	Kaohsiung City	26	Good	0.6	0.15	28.5	12.8	26	6	2.4	3.5	1	1	152	0.1	
01/08/2024 11:00:00 AM	Magong	Penghu County	20	Good	0.6	0.15	28.3	22.2	10	6	4.3	6.3	1.9	1.9	175	0.1	
01/08/2024 10:00:00 AM	Meinong	Kaohsiung City	19	Good	0.6	0.15	24.8	9.3	19	6	1.7	3.3	1.5	1.5	252	0.1	
31/07/2024 11:00:00 PM	Pingtung (Liuqiu)	Pingtung County	28	Good	0.6	0.15	13.6	17.1	18	6	2	2.5	0.4	0.3	167	0.1	
28/07/2024 4:00:00 AM	Wanli	New Taipei City	26	Good	0.6	0.15	8.1	9.7	26	6	1.9	3.3	1.4	1.3	197	0.1	
22/07/2024 6:00:00 PM	Longtan	Taoyuan City	22	Good	0.6	0.15	17.3	24.2	20	6	5.9	8.1	2.2	4.1	64	0.1	
22/07/2024 2:00:00 AM	Fengyuan	Taichung City	25	Good	0.6	0.15	9.8	20.9	6	6	6.4	7.3	0.9	1.6	114	0.1	
21/07/2024 7:00:00 PM	Zhongming	Taichung City	35	Good	0.6	0.15	28.6	38	11	6	4.4	5.9	1.4	2.6	164	0.1	
21/07/2024 1:00:00 AM	Tainan (Madou)	Tainan City	21	Good	0.6	0.15	12.4	27.5	22	6	4.6	5.9	1.2	0.2	120	0.1	

Search

cleaned data

Σ aqi

Σ co

Σ co_8hr

country

>

date

Σ latitude

Σ longitude

Σ no

Σ no2

Σ nox

Σ o3

Σ o3_8hr

Σ pm10

Σ pm10_avg

Σ pm2_5

Σ pm2_5_avg

Σ siteid

sitename

Σ so2

Σ so2_avg

status

Σ winddirec

Σ windspeed

• Data Preparation

pm10_avg	so2_avg	longitude	latitude	siteid
17	0	120.28676111	22.63390278	16567
17	0	121.08895493	24.74091408	17853
19	0	120.677306	23.756389	19860
18	0	120.24583056	23.17904722	20458
11	0	121.08895493	24.74091408	20504
15	0	121.203986	24.952786	20517
19	0	121.203986	24.952786	20677
20	0	121.203986	24.952786	20762
15	0	120.568794	24.225628	25274
13	0	121.689881	25.179667	27231
9	0	121.36548982	25.07798949	29446
2	0	120.305689	22.757506	29908
25	0	121.599769	23.971306	30002
12	0	119.952724	26.153736	30991
9	0	120.348742	23.757547	32912
16	0	120.685306	23.913	35461
20	0	121.15045	22.755358	40168
14	0	121.529583	25.182722	40181
17	0	120.568794	24.225628	40376
11	0	120.75956754	24.38248443	44907
11	0	120.75956754	24.38248443	45161
16	0	121.15045	22.755358	50228
17	0	120.312017	22.608711	57404
26	0	120.530542	22.883583	58053
10	0	119.566158	23.569031	62009
19	0	120.530542	22.883583	62111
18	0	120.37722	22.35222	62982
26	0	121.689881	25.179667	70699
18	0	121.21645772	24.86400048	81759



• Data Preparation

= Table.RenameColumns("#Changed Type",{{"county", "country"}})							
	date	sitename	country	aqi	status	so2	co
1	31/08/2024 11:00:00 PM	Hukou	Hsinchu County	62	Moderate	0.9	
2	31/08/2024 11:00:00 PM	Zhongming	Taichung City	50	Good	1.6	
3	31/08/2024 11:00:00 PM	Zhudong	Hsinchu County	45	Good	0.4	
4	31/08/2024 11:00:00 PM	Hsinchu	Hsinchu City	42	Good	0.8	
5	31/08/2024 11:00:00 PM	Toufen	Miaoli County	50	Good	1	
6	31/08/2024 11:00:00 PM	Miaoli	Miaoli County	40	Good	1.1	
7	31/08/2024 11:00:00 PM	Sanyi	Miaoli County	39	Good	0.9	
8	31/08/2024 11:00:00 PM	Fengyuan	Taichung City	44	Good	1.3	
9	31/08/2024 11:00:00 PM	Shalu	Taichung City	46	Good	2.5	
10	31/08/2024 11:00:00 PM	Dali	Taichung City	49	Good	0.7	
11	31/08/2024 11:00:00 PM	Xitun	Taichung City	44	Good	1.9	
12	31/08/2024 11:00:00 PM	Pingzhen	Taoyuan City	58	Moderate	0.1	
13	31/08/2024 11:00:00 PM	Xianxi	Changhua County	45	Good	0.9	
14	31/08/2024 11:00:00 PM	Erlin	Changhua County	62	Moderate	0.2	
15	31/08/2024 11:00:00 PM	Nantou	Nantou County	58	Moderate	0.9	
16	31/08/2024 11:00:00 PM	Douliu	Yunlin County	61	Moderate	1	
17	31/08/2024 11:00:00 PM	Lunbei	Yunlin County	63	Moderate	0.3	
18	31/08/2024 11:00:00 PM	Xingang	Chiayi County	43	Good	1.5	
19	31/08/2024 11:00:00 PM	Puzi	Chiayi County	41	Good	1.4	
20	31/08/2024 11:00:00 PM	Taixi	Yunlin County	44	Good	0.6	
21	31/08/2024 11:00:00 PM	Longtan	Taoyuan City	60	Moderate	1.8	
22	31/08/2024 11:00:00 PM	Guanyin	Taoyuan City	65	Moderate	1.4	
23	31/08/2024 11:00:00 PM	Xinying	Tainan City	38	Good	0.8	
24	31/08/2024 11:00:00 PM	Cailiao	New Taipei City	65	Moderate	0.9	
25	31/08/2024 11:00:00 PM	Pingtung (Fangshan)	Pingtung County	13	Good	0.3	
26	31/08/2024 11:00:00 PM	Keelung	Keelung City	58	Moderate	1.5	
27	31/08/2024 11:00:00 PM	Xizhi	New Taipei City	66	Moderate	0.6	
28	31/08/2024 11:00:00 PM	Wanli	New Taipei City	59	Moderate	0.3	
29	31/08/2024 11:00:00 PM	Xindian	New Taipei City	66	Moderate	1.1	
30	31/08/2024 11:00:00 PM	Tucheng	New Taipei City	61	Moderate	0.4	
31	31/08/2024 11:00:00 PM	Banqiao	New Taipei City	57	Moderate	2.1	
32	31/08/2024 11:00:00 PM	Xinzhuang	New Taipei City	64	Moderate	1.5	

Query Settings

PROPERTIES

Name

cleaned data

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Renamed Columns

• Data Preparation

1.2 windspeed	1 ² ₃ winddirec	1.2 co_8hr	1.2 pm2_5_avg	1 ² ₃ pm10_avg	1 ² ₃ so2_avg
2.3	225	0.2	20.1	26	1
1.1	184	0.2	15.3	23	1
0.4	210	0.2	13.8	24	0
1.9	239	0.2	13	26	1
1.8	259	0.1	15.3	28	1
1.4	235	0.1	12.2	17	1
0.6	203	0.1	11.4	16	0
1.2	38	0.2	13.5	21	1
0.5	8	0.1	14.2	26	1
0.9	97	0.2	15.1	21	0
0.9	244	0.1	13.6	26	1
1.2	252	0.3	18.2	32	0
0.3	19	0.1	14	25	1
0.6	65	0.1	20	32	0
1.6	164	0.2	18.5	29	1
1	69	0.1	19.6	35	1
0.3	84	0.1	20.5	30	0
0.5	220	0.1	13.1	18	1
0.4	90	0.2	12.6	23	1
1.3	31	0.1	13.7	22	0
2.9	217	0.2	19.2	31	2
1.2	224	0.1	21.3	32	1
0.3	110	0.2	11.7	19	0
0.7	194	0.3	21.1	26	1
3.8	96	0	2.4	13	0
0.7	324	0.2	18.5	26	2
1	266	0.2	21.7	28	1
1.7	217	0.2	18.7	33	1
1.6	195	0.2	21.5	27	1
0.8	156	0.3	19.7	29	1
1.2	256	0.3	17.8	27	2
1	240	0.3	20.7	34	1



• Data Preparation

1.2 o3	1.2 o3_8hr	1 ² ₃ pm10	1 ² ₃ pm2_5	1.2 no2	1.2 nox	1.2 no
35	40.2	18	17	2.3	2.6	
27.9	35.1	27	14	7.6	9.3	
25.1	40.6	21	13	2.9	4.1	
30	35.9	19	10	4	4.8	
33.5	35.9	18	14	1.8	3.1	
35.2	35	15	12	4	5.1	
35.3	42.9	14	9	2.4	3.1	
24.6	39.7	21	12	6.8	7.3	
30.3	40.4	33	16	7.3	7.7	
29.4	37	20	12	5.6	6.3	
28.5	38	32	17	7.3	7.7	
23.3	38.5	29	18	9.8	10.5	
41.4	46.5	34	12	3.5	3.8	
31.2	41.9	37	21	2.7	3.8	
35.6	40.7	17	13	3.7	4.7	
30.5	39	31	19	5.3	6	
32	35.7	37	29	2.1	3.6	
33.5	41	30	18	4.2	4.5	
35.5	40.7	25	19	6.6	7.7	
36.2	40.7	24	15	2.8	4	
26.9	42.3	26	20	4.6	5.2	
30.6	38.7	24	17	8	8.3	
28.9	41.6	21	13	10.5	12	
36.5	50.8	19	15	10.7	12	
15.1	14.9	15	4	0.9	1.4	
28	57.9	22	16	9.6	10	
23.5	50.7	25	21	14.3	16	
25.5	46	26	19	4.6	5.3	
25.6	46.6	32	19	8.9	9.6	
18.5	45	31	20	14.7	15.3	
25.6	48.7	27	18	14.3	16	
29.2	48.3	32	25	11.8	12.4	

• Data Preparation

1.2 longitude	1.2 latitude	1.2 siteid
121.0388689	24.90009696	1
120.641092	24.151958	2
121.0889549	24.74091408	3
120.9723675	24.8056356	4
120.8986929	24.69690679	5
120.8201147	24.56499183	6
120.7595675	24.38248443	7
120.7425241	24.25699731	8
120.568794	24.225628	9
120.6784444	24.09961111	10
120.616917	24.162197	11
121.203986	24.952786	12
120.469061	24.131672	13
120.409653	23.925175	14
120.685306	23.913	15
120.544994	23.711853	16
120.348742	23.757547	17
120.345531	23.554839	18
120.2478	23.46538	19
120.1993333	23.702175	20
121.2164577	24.86400048	21
121.0828309	25.03556747	22
120.31725	23.305633	23
121.481028	25.06895	24
120.651472	22.260899	25
121.760056	25.129167	26
121.64081	25.06624	27
121.689881	25.179667	28
121.537778	24.977222	29
121.451861	24.982528	30
121.458667	25.012972	31
121.4325	25.037972	32



• Data Visualization

So after we clean data we have this visualizations that explains each thing we have and answers some important business questions.



Air Quality Analysis Dashboard

Dashboard

Slicers and Decomposition Tree

Business Charts

Spatial Map Analysis

Average AQI

53.68

Average Pm2.5

16.31

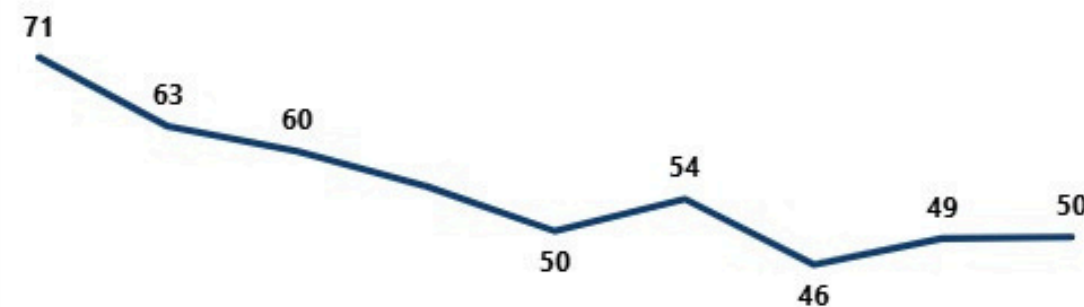
Count AQI

5.9M

Average CO

0.32

Average of AQI Overtime



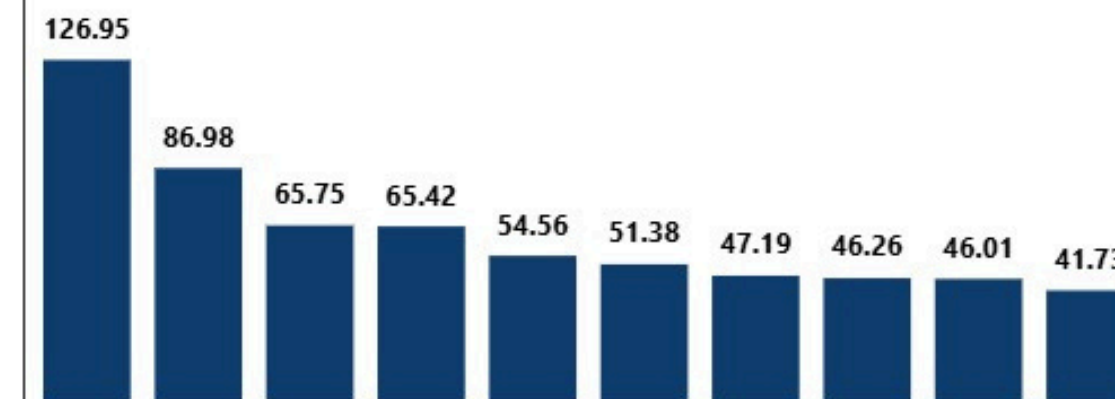
Average of AQI By Country



Average of AQI By Month



Top 10 Monitoring Stations By Avg AQI



• Data Visualization



Dashboard

Slicers and Decomposition Tree

Business Charts

Spatial Map Analysis

Year

All

Status

All

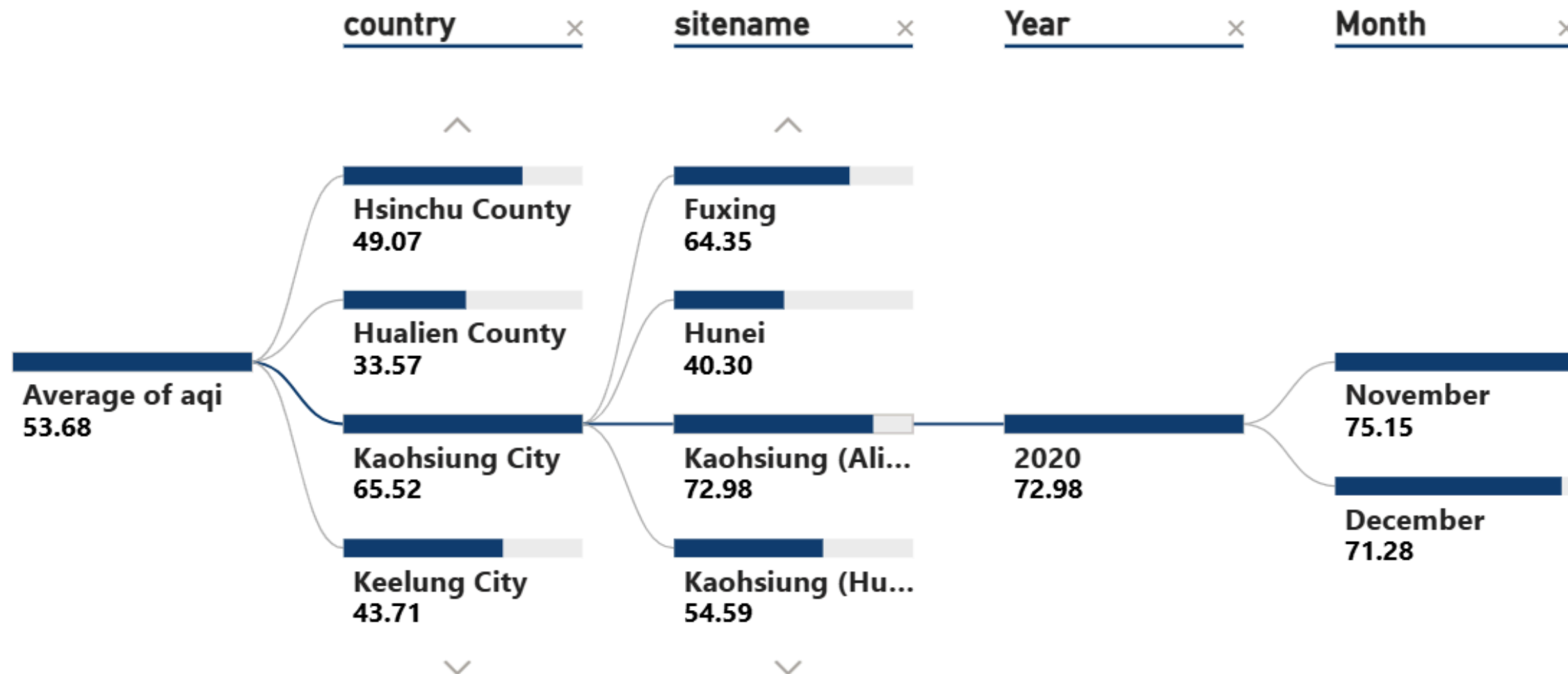
Country

All

Site Name

All

Average AQI Breakdown by Location and Time



• Data Visualization

Dashboard

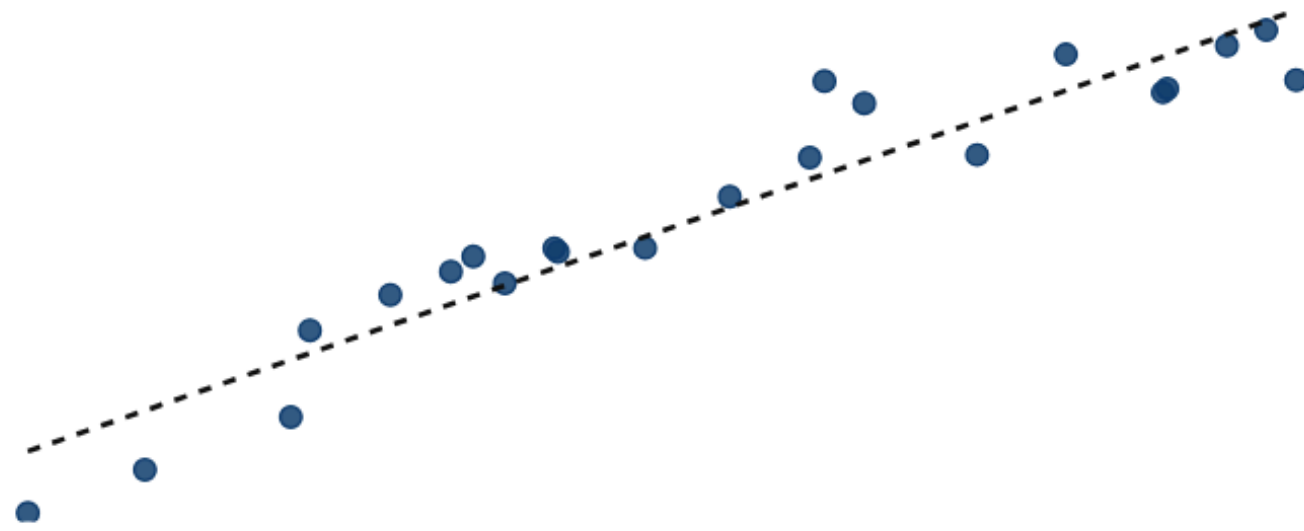
Slicers and Decomposition Tree

Business Charts

Spatial Map Analysis

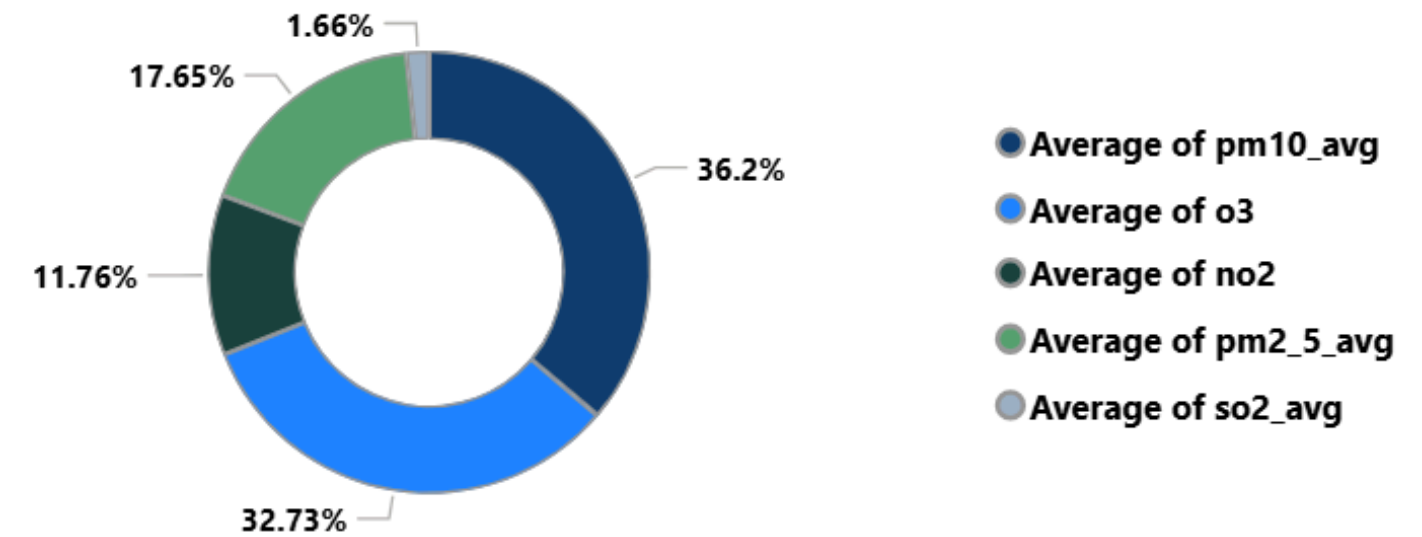
Is there a direct statistical correlation between particulate matter (PM₁₀) and overall AQI levels?

AQI Correlation with PM10



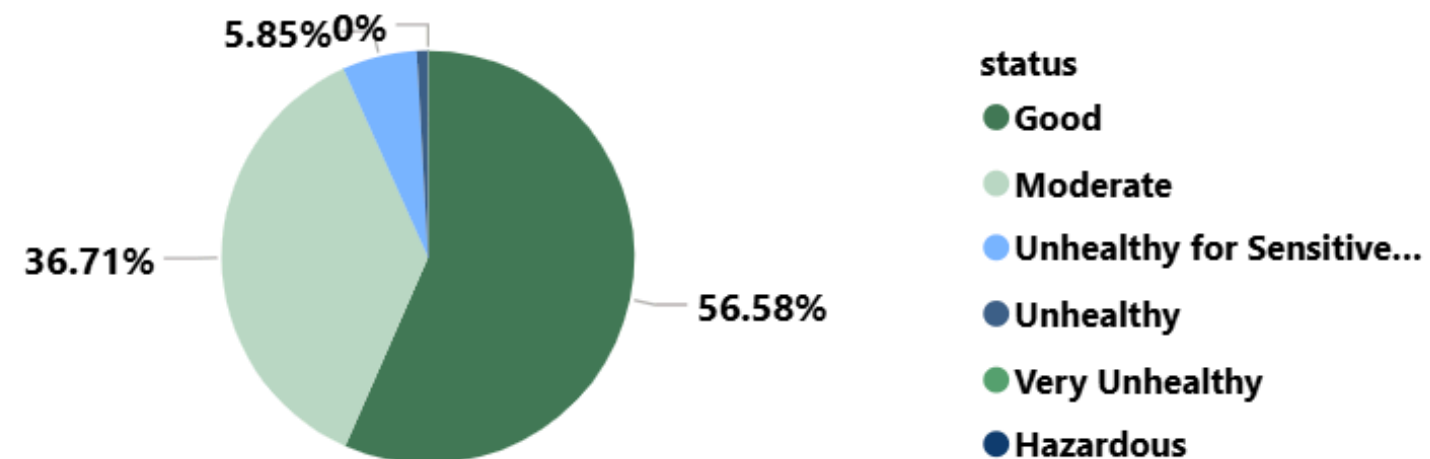
Which primary pollutant drives the majority of air pollution?

Pollutant Contribution Profile (By Average)



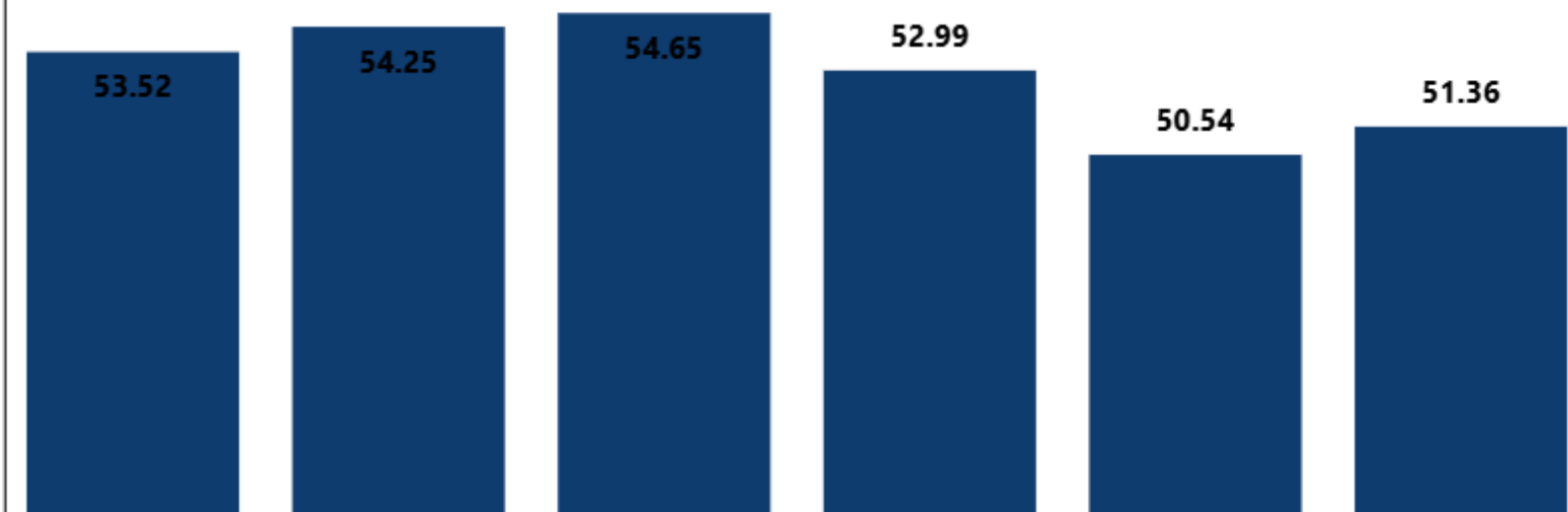
What proportion of the year offers healthy vs. hazardous air quality for the public?

AQI Status Proportions

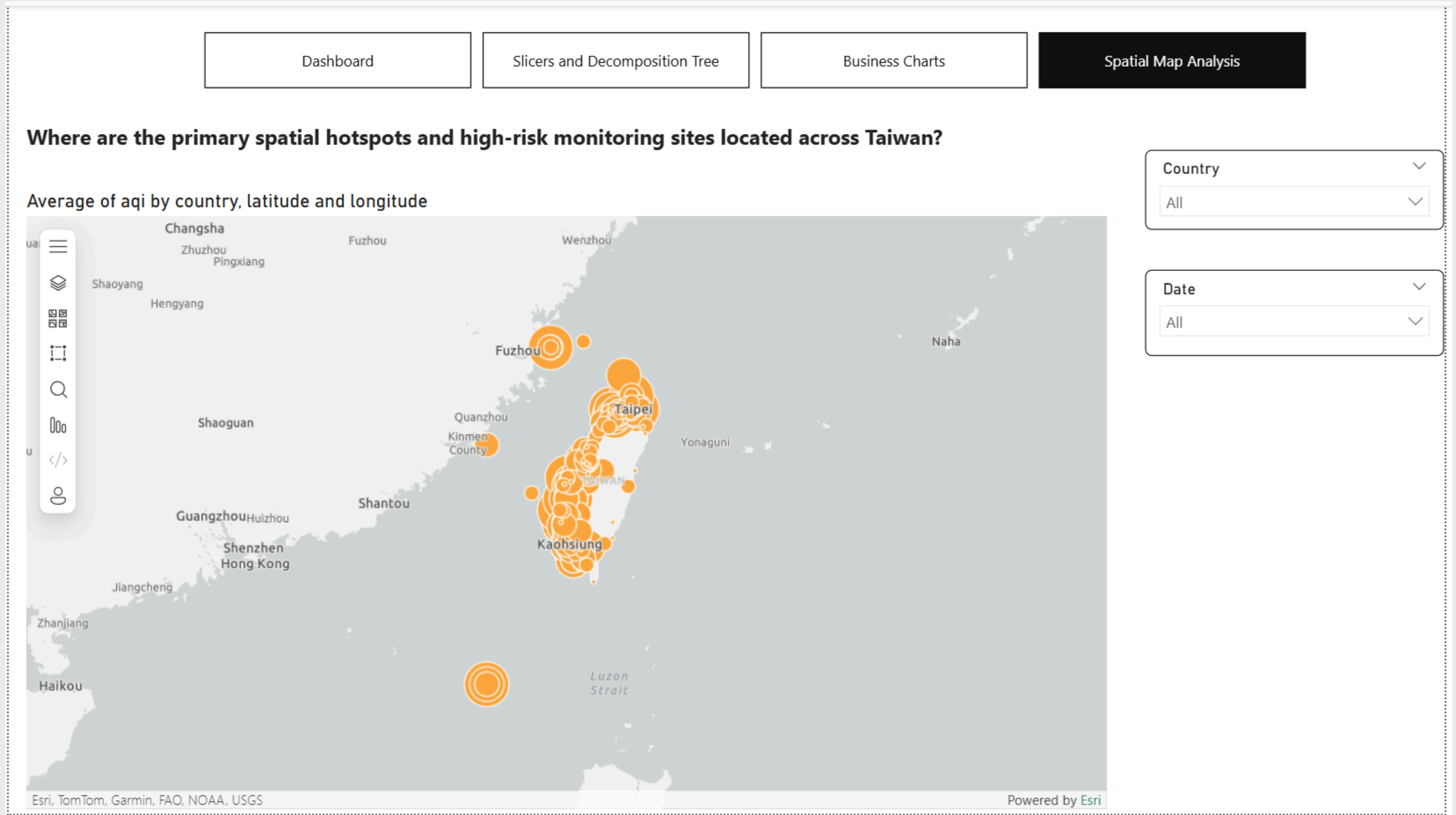


How do meteorological factors like wind speed influence pollution dispersion?

Impact of Wind Speed Bins on Average AQI



• Data Visualization



• Data Modeling & Evaluation

Modeling(Darin Ashraf - Mariam Gamal)

```
models = {  
    "Linear Regression":  
        LinearRegression(),  
    "Decision Tree":  
        DecisionTreeRegressor(  
            random_state=42  
        )  
}
```

```
results = []  
  
trained_models = {}  
  
for name, model in models.items():  
  
    model.fit(X_train, y_train)  
  
    prediction = model.predict(X_test)  
  
    mae = mean_absolute_error(  
        y_test,  
        prediction  
    )
```

```
    rmse = np.sqrt(  
        mean_squared_error(  
            y_test,  
            prediction  
        )  
    )  
  
    r2 = r2_score(  
        y_test,  
        prediction  
    )  
  
    results.append({  
  
        "Model": name,  
  
        "MAE": mae,  
  
        "RMSE": rmse,  
  
        "R2 Score": r2  
  
    })  
  
    trained_models[name] = model  
  
results = pd.DataFrame(results)  
  
results = results.sort_values(  
    by="R2 Score",
```



• Data Modeling & Evaluation

```
ascending=False
)

results
```

	Model	MAE	RMSE	R2 Score
1	Decision Tree	0.155050	1.137115	0.994353
0	Linear Regression	3.383254	4.818046	0.898619

Next steps: [Generate code with results](#) [New interactive sheet](#)

```
best_model_name = results.iloc[0]["Model"]

print("Best Model :", best_model_name)

best_model = trained_models[best_model_name]
```

Best Model : Decision Tree

```
prediction = best_model.predict(X_test)

comparison = pd.DataFrame({
    "Actual": y_test,
```

```
"Predicted": prediction
})

comparison.head()
```

	Actual	Predicted
2476	49.0	49.0
6771	47.0	47.0
6855	56.0	56.0
3047	28.0	28.0
4092	23.0	23.0

Next steps: [Generate code with comparison](#) [New interactive sheet](#)

```
plot_df = pd.DataFrame({
    "actual": y_test.values,
    "predicted": prediction
})

plot_df = plot_df.sort_values("actual")

plot_df = plot_df.iloc[
    np.linspace(0, len(plot_df) - 1, 1000).astype(int)
]
```



• Data Modeling & Evaluation

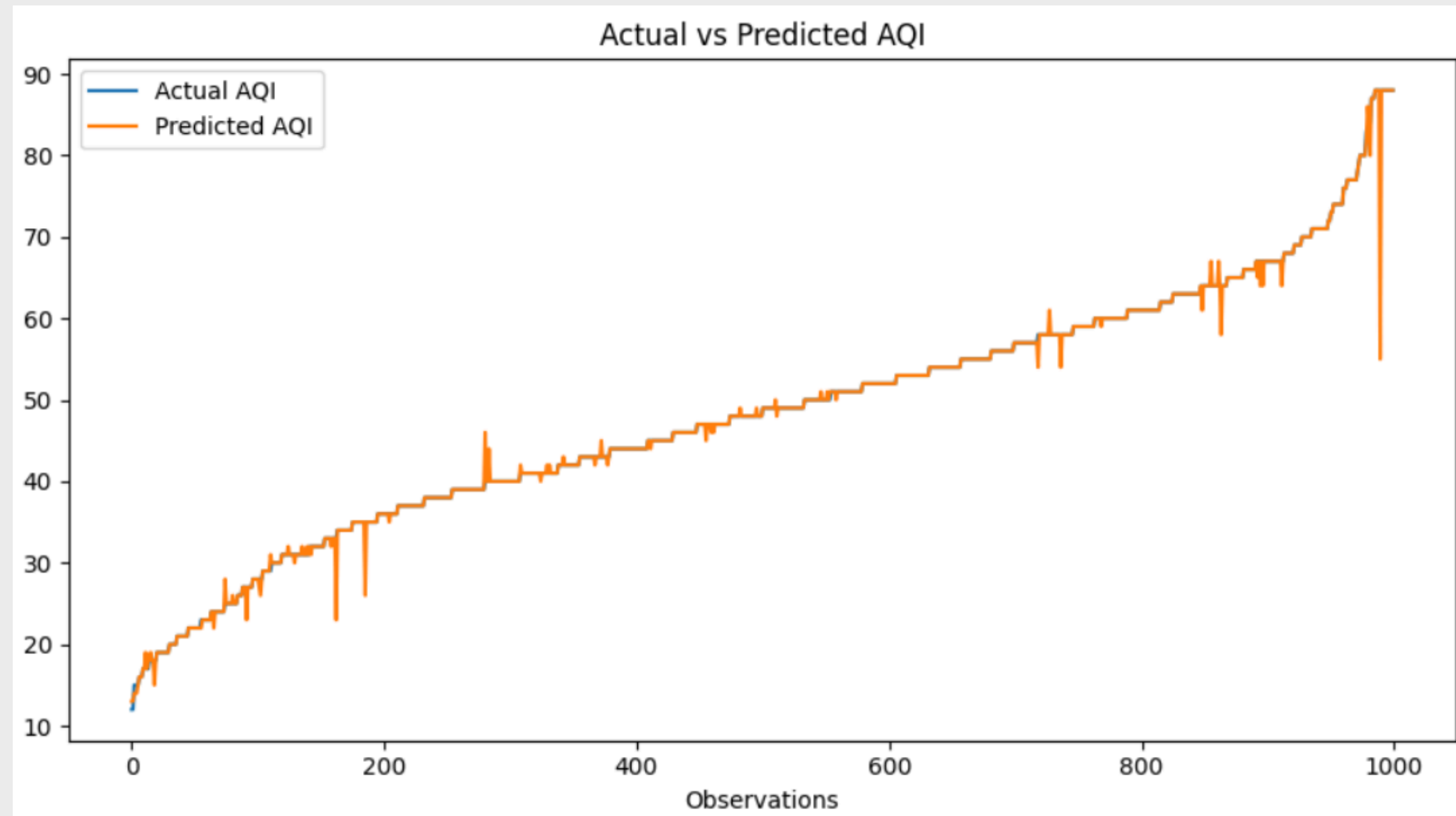
```
plt.figure(figsize=(9, 5))

plt.plot(
    plot_df["actual"].values,
    label="Actual AQI"
)

plt.plot(
    plot_df["predicted"].values,
    label="Predicted AQI"
)

plt.xlabel("Observations")
plt.ylabel("AQI")
plt.title("Actual vs Predicted AQI")
plt.legend()

plt.tight_layout()
plt.show()
```



• Data Modeling & Evaluation

```
eval_df = pd.DataFrame({
    "actual": y_test.values,
    "predicted": prediction
})

eval_df["error"] = abs(
    eval_df["actual"] - eval_df["predicted"]
)

bins = [0, 25, 50, 75, 100, 127]

labels = [
    "0-25",
    "26-50",
    "51-75",
    "76-100",
    "101-127"
]

eval_df["aqi_range"] = pd.cut(
    eval_df["actual"],
    bins=bins,
    labels=labels,
    include_lowest=True
)

mae_by_range = (
```

```
    mae_by_range = (
        eval_df
        .groupby("aqi_range", observed=False)["error"]
        .mean()
    )

plt.figure(figsize=(8, 5))

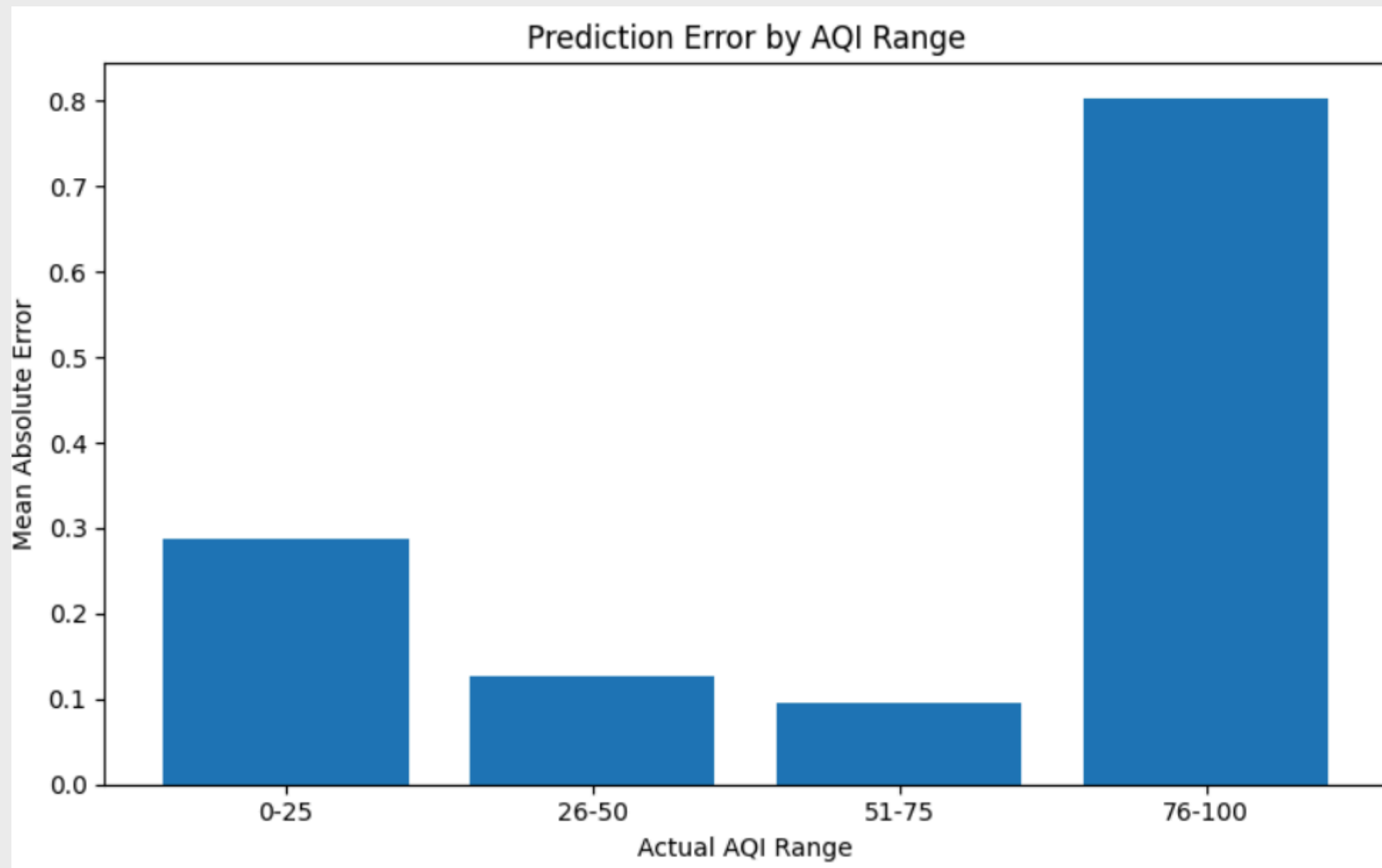
plt.bar(
    mae_by_range.index.astype(str),
    mae_by_range.values
)

plt.xlabel("Actual AQI Range")
plt.ylabel("Mean Absolute Error")
plt.title("Prediction Error by AQI Range")

plt.tight_layout()
plt.show()
```



• Data Modeling & Evaluation



```
residuals = prediction - y_test

plt.figure(figsize=(8, 5))

plt.hist(
    residuals,
    bins=50
)

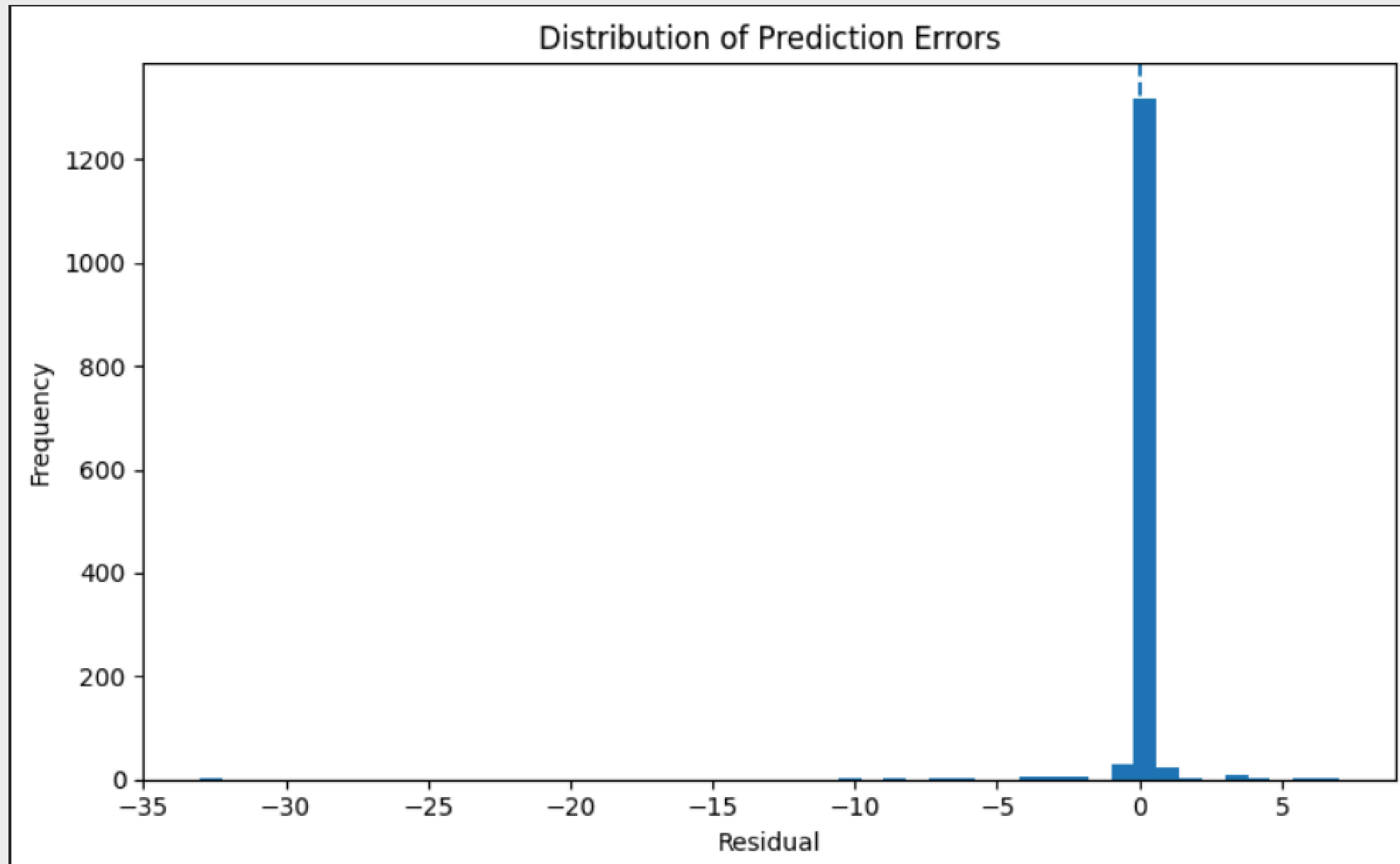
plt.axvline(
    0,
    linestyle="--"
)

plt.xlabel("Residual")
plt.ylabel("Frequency")
plt.title("Distribution of Prediction Errors")

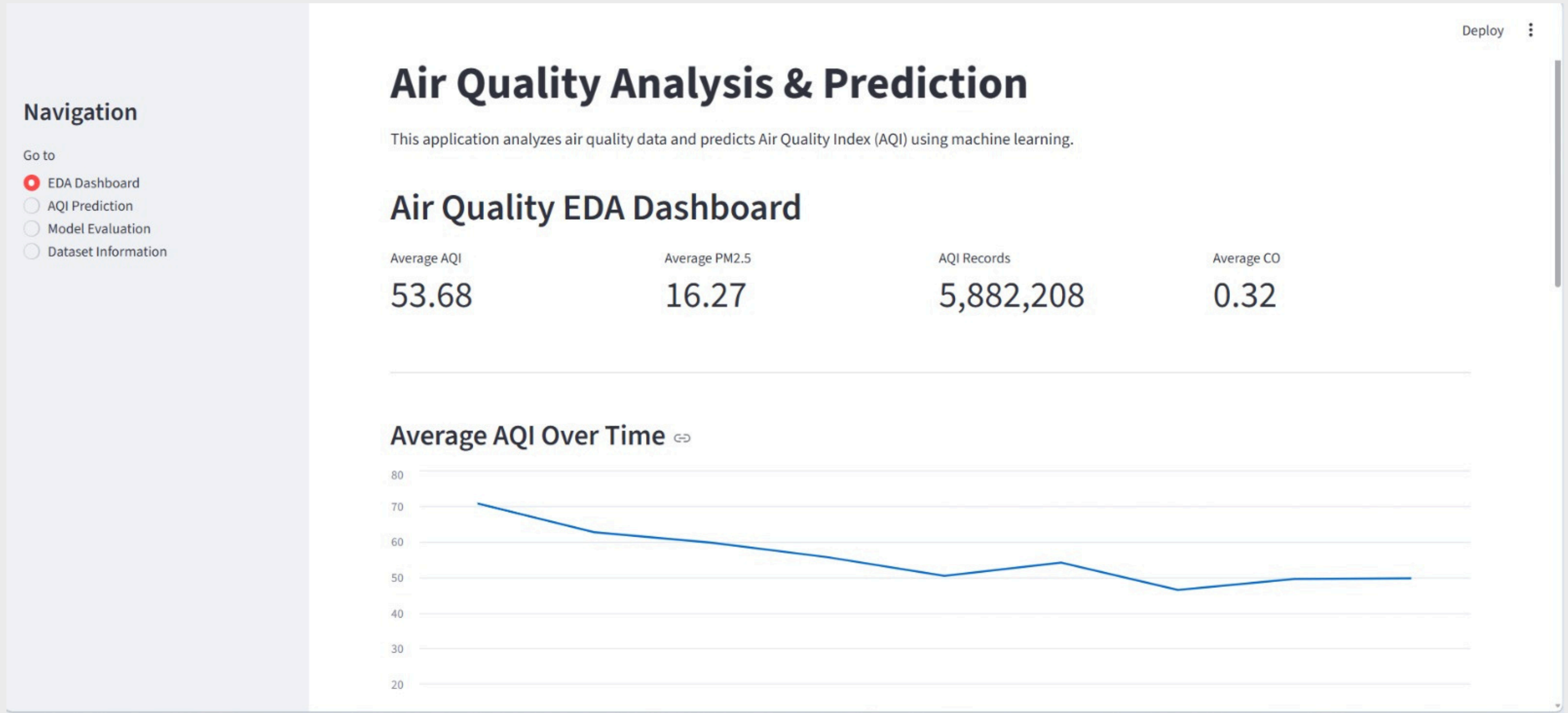
plt.tight_layout()
plt.show()
```



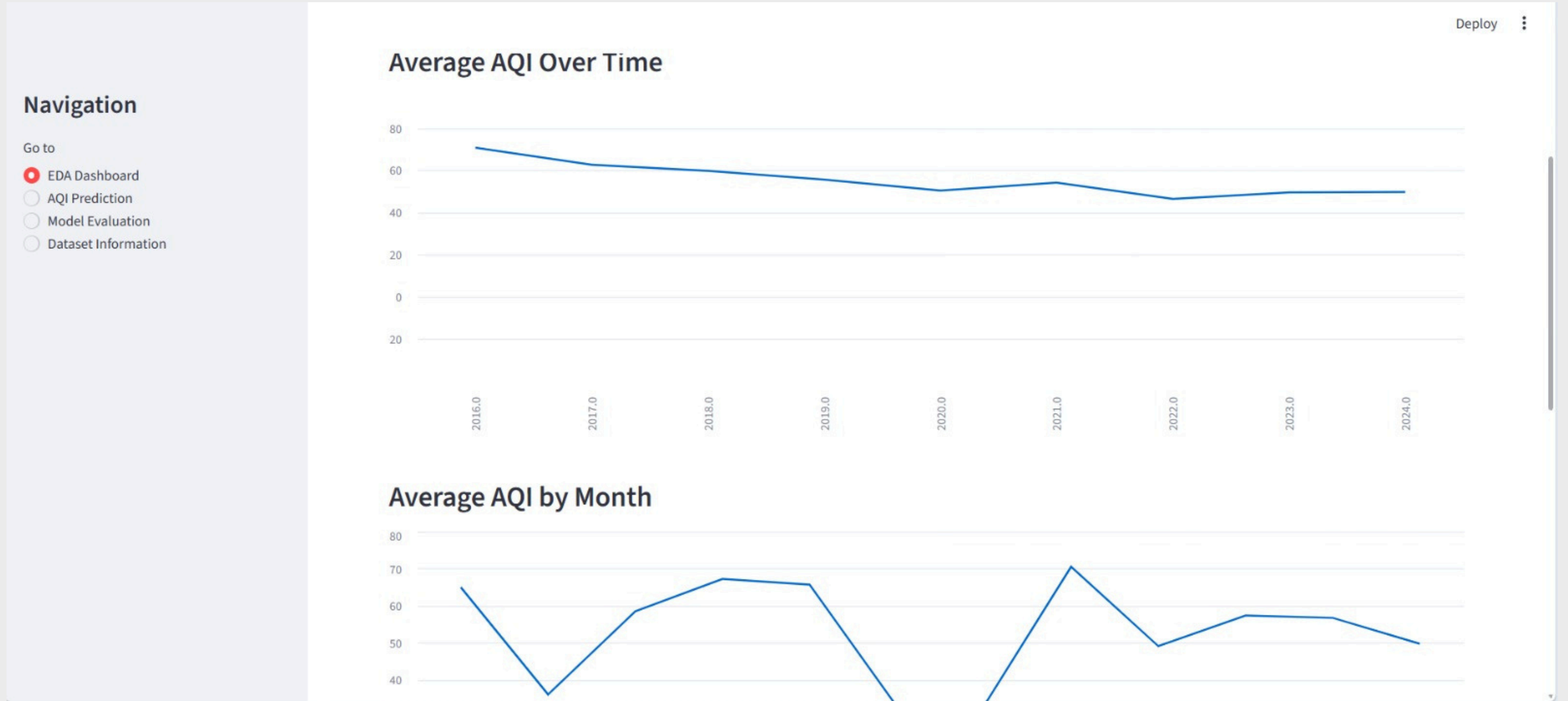
• Data Modeling & Evaluation



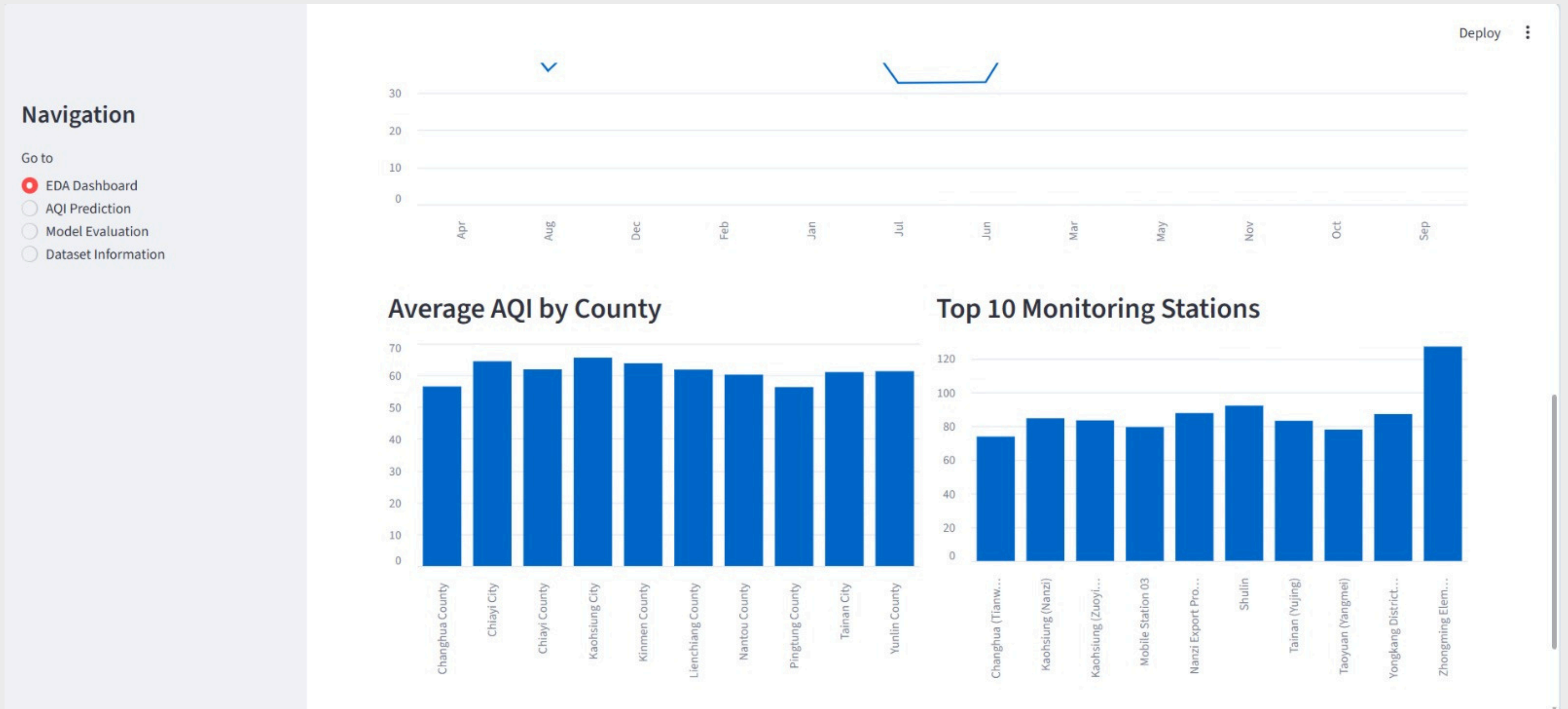
• Streamlit app



• Streamlit app



• Streamlit app



• Streamlit app

Navigation

Go to

- ☐ EDA Dashboard
- ☒ AQI Prediction
- ☐ Model Evaluation
- ☐ Dataset Information

Deploy

Enter the environmental and pollutant measurements to predict the AQI.

SO2	CO	O3
1.70 - +	0.29 - +	28.30 - +
O3_8HR	PM10	PM2_5
29.00 - +	28.00 - +	14.00 - +
NO2	NOX	NO
9.00 - +	10.80 - +	1.40 - +
WINDSPEED	WINDDIREC	CO_8HR
1.70 - +	137.00 - +	0.30 - +
PM2_5_AVG	PM10_AVG	SO2_AVG
14.00 - +	29.00 - +	1.00 - +

Predict AQI

Prediction completed successfully!



• Streamlit app

Navigation

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Deploy 

PM2_5_AVG

14.00 - +

PM10_AVG

29.00 - +

SO2_AVG

1.00 - +

Predict AQI

Prediction completed successfully!

Predicted AQI

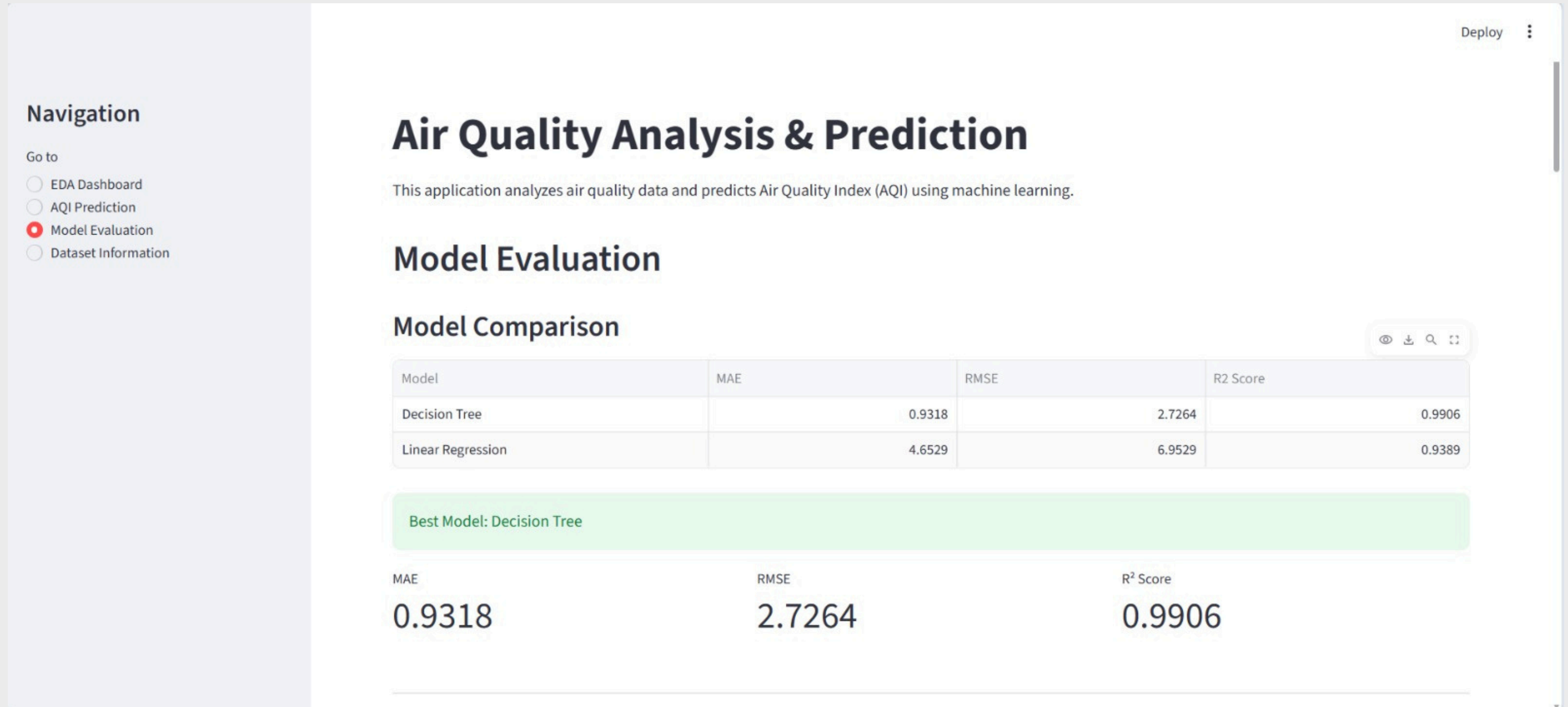
47.00

AQI Category: **Good**

Model used: **Decision Tree**



• Streamlit app



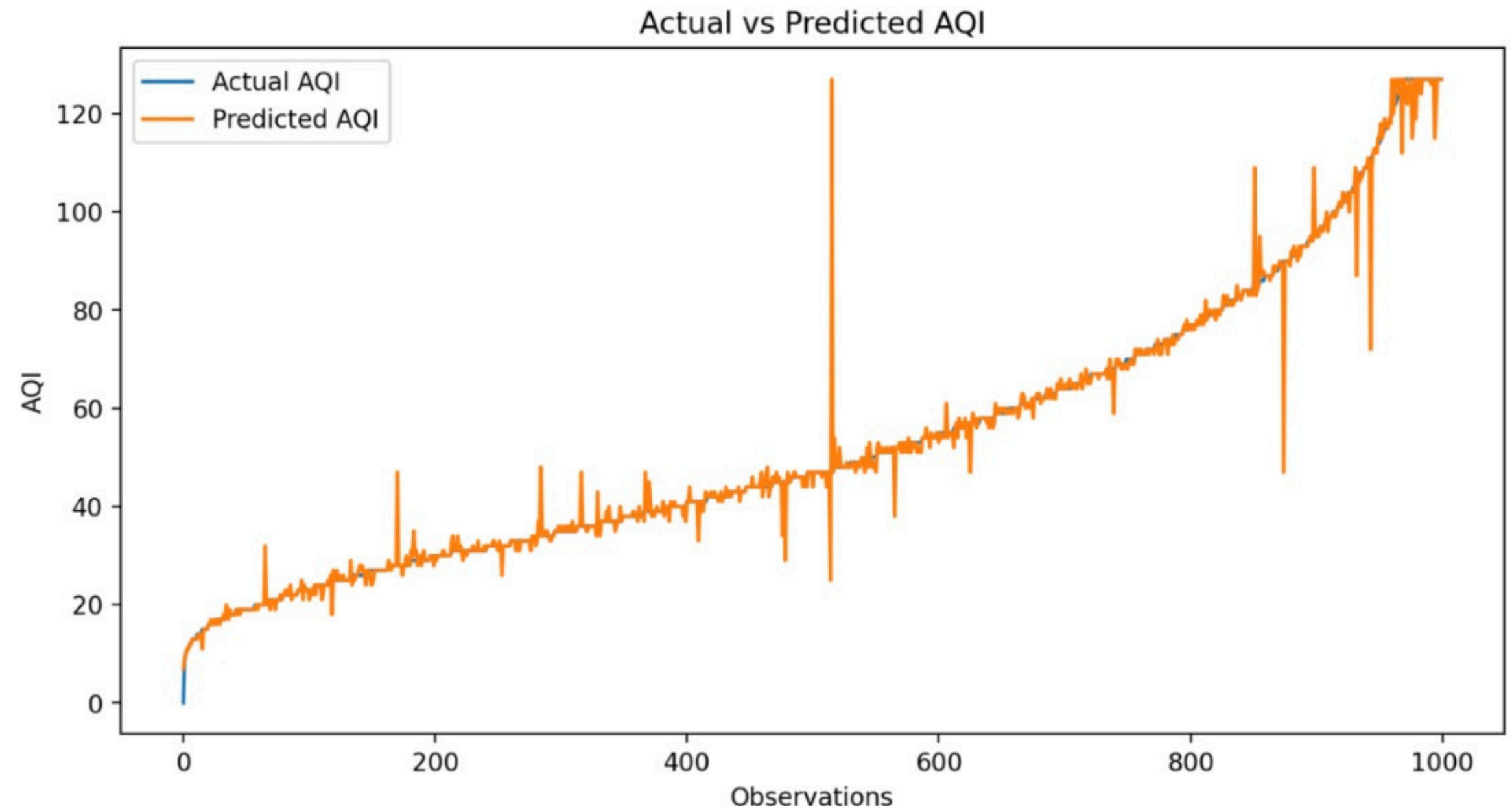
• Streamlit app

Navigation

Go to

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Actual vs Predicted AQI



Deploy



• Streamlit app

Navigation

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☐ EDA Dashboard

☐ AQI Prediction

☐ Model Evaluation

☒ Dataset Information

Deploy

Air Quality Analysis & Prediction

This application analyzes air quality data and predicts Air Quality Index (AQI) using machine learning.

Dataset Information

Rows

5,882,208

Columns

23

Duplicates

0

Dataset Preview

	date	sitename	county	aqi	status	so2	co	o3	o3_8hr	pm10	pm2_5	no2	nox	no	windspeed	v
90	2024-08-31 22:00:00	Toufen	Miaoli County	52	Moderate	1.2	0.17	32.4	36.6	33	17	3.4	4.8	1.3	2.6	
91	2024-08-31 22:00:00	Hsinchu	Hsinchu City	44	Good	0.7	0.25	30	38.2	23	10	6.7	7.5	0.7	2.3	
92	2024-08-31 22:00:00	Zhongming	Taichung City	52	Moderate	1	0.25	31	38.6	22	20	7	8.7	1.7	0.6	
93	2024-08-31 22:00:00	Longtan	Taoyuan City	60	Moderate	1.7	0.2	27.1	48.6	25	14	6.3	7	0.6	4.3	
94	2024-08-31 22:00:00	Dali	Taichung City	53	Moderate	1.2	0.27	27.7	40.7	29	16	7.2	8.2	0.9	1.8	

• Streamlit app

Navigation

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Deploy

98	2024-08-31 22:00:00	Xianxi	Changhua County	46	Good	1.3	0.25	41.7	47.8	34	18	4	4.3	0.2	0.3
99	2024-08-31 22:00:00	Nantou	Nantou County	61	Moderate	1.4	0.22	33.4	44.5	32	18	5.6	6.7	1	1.3

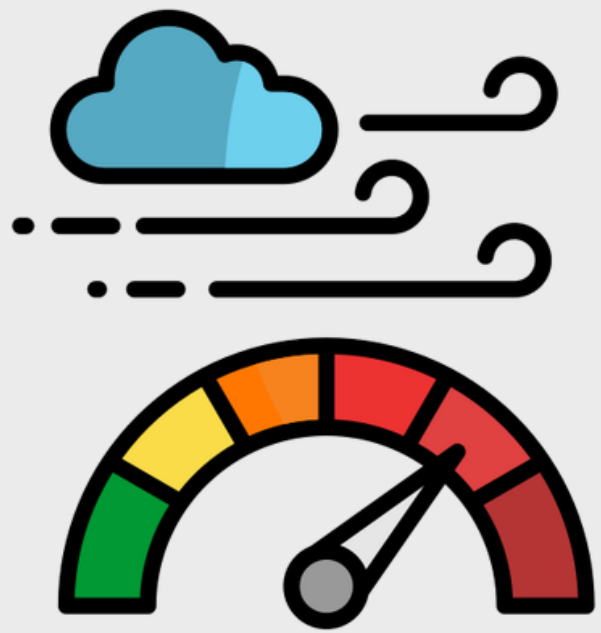
Descriptive Statistics

	pm10	pm2_5	no2	nox	no	windspeed	winddirec	co_8hr	pm2_5_avg	pm10_avg	so2_avg	longitude	latitude	siteid
count	5882208	5882208	5882208	5882208	5882208	5882208	5882208	5882208	5882208	5882208	5882208	4948588	4948588	5882208
mean	33.2536	16.2702	10.869	13.1325	1.9486	2.0669	156.6204	0.3036	16.3071	33.4549	1.5061	120.8235	24.045	2941104.5
std	20.3702	10.8623	7.5353	9.009	1.6085	1.3522	116.1703	0.167	10.4604	19.6858	1.0306	0.9644	1.002	1698047.3305
min	0	0	0	0	0	0	0	0	0	0	0	0	0	1
25%	18	8	5.2	6.4	0.8	1.1	46	0.2	8	19	1	120.3455	23.0482	1470552.75
50%	28	14	9	10.8	1.4	1.7	137	0.3	14	29	1	120.6853	24.152	2941104.5
75%	44	22	15	17.5	2.6	2.8	265	0.4	22	45	2	121.4492	25.013	4411656.25
max	83	43	29.7	34.15	5.3	5.35	593.5	0.7	43	84	3.5	121.8157	26.258	5882208

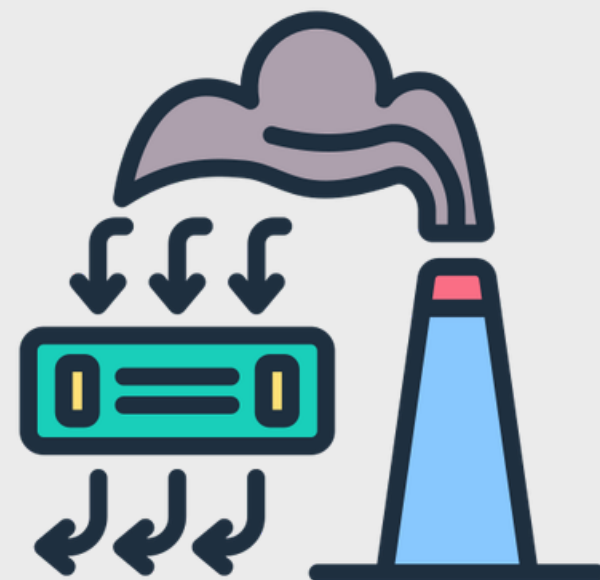
• Conclusion

- This project provided a complete data-driven analysis of air quality in Taiwan. The dataset was cleaned and prepared to ensure reliable results, followed by exploratory analysis to identify pollution patterns and relationships between air quality and different pollutants.
- Machine Learning models were then used to predict AQI and evaluate their performance using appropriate regression metrics. The final solution provides useful insights into air pollution and demonstrates how data analysis and machine learning can support better air-quality monitoring and prediction.





Thank you



- **Team Members**

- Aseel Waleed
- Dareen Ashraf
- Cinderella Ehab
- Mariam Gamal

